When Does Adapting the Meta-Parameter Pay? Minimal Governors and Structural Limits for Weight- and Horizon-Governed MPC

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***Abstract*—A slow outer loop that retunes an inner model predictive controller — its scalarization weight, or its prediction horizon — is an appealing idea, and we set out to make an elaborate version of it work: a second-order wave field on the interconnection graph, transporting each subsystem's difficulty to its neighbours. It lost in every physically grounded arena we built. What replaces it is sharper than a negative result. First, two minimal governors that do pay. In a chain with a physically travelling disturbance, an *affine* delay line — per-hop delay minus a constant anticipation term — recovers 91% of a clairvoyant bound and cuts closed-loop cost by 12.4% against the best possible local detector; the field, a shaped filter, and the proportional delay family that most designs default to are all significantly worse, and the missing constant term is precisely why. In a fleet of loops sharing one embedded controller under a hard computation budget, a governor that requests the maximum horizon *only* when the relaxed-dynamic-programming certificate breaks, and decays slowly otherwise, beats the best fixed horizon at equal *spent* computation by 5–6% across budgets, for independent and for perfectly correlated disturbances alike; a decomposition shows the temporal mechanism pays always, while cross-loop allocation pays only when disturbances are independent. Second, structural limits that explain why nothing more elaborate could win: amplitude-scale invariance of the optimal meta-parameter, the linear-time-invariance of the graph field, the DC cross-gain of its forcing, and a quantization-limited reach law. Third, a sign correction to the multiparametric machinery of a widely used multiobjective MPC formulation: its weighted value function is concave, not convex, in the weights, so its weight-selection program is a safe inner approximation rather than an equivalent one — stated, proved, and repaired here. Fourth, the part most likely to be useful beyond our own problem: five experimental leaks, two acausal, two of accounting, one of channel, every one of which inverted a conclusion of ours before adversarial review caught it, together with the cheap executable tests that now keep each of them out.**

***Index Terms*—Predictive control, adaptive horizon, resource-aware control, distributed control, experimental methodology, negative results.**

I. Introduction

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Every numerical claim below is regenerated by archived scripts. The defects found during four rounds of adversarial review of this work — five of them serious enough to invert a conclusion — are each encoded as a regression assertion in an executable audit suite distributed with the code: https://github.com/fmarrabal/limits-horizon-governed-mpc.

MODEL predictive control exposes a handful of design quantities that are usually chosen once and then frozen: how much one objective matters relative to another, how far ahead to predict, how hard to push soft constraints. Letting these *meta-parameters* vary online, driven by some measured indicator of how hard the current situation is, is a natural thing to want, and adaptive-horizon MPC, resource-aware MPC and priority scheduling all do a version of it. A distributed version suggests itself just as readily: couple the meta-parameters of neighbouring subsystems through a diffusion or wave operator on the interconnection graph, so that one subsystem's decision reaches the others without anyone having to negotiate.

We set out to make that work, across six arenas built for the purpose. The elaborate governor won in exactly one of them — a synthetic bench whose priority signal had no physical origin, which is why we do not count it — and in none of the physically grounded ones. This paper reports what survived instead, and it turned out to be more useful than the idea we started with.

The first thing that survived is a pair of *minimal* governors that genuinely pay, each beating not only its elaborate rivals but the natural simple ones, and each for a reason that can be stated in one sentence. In a transport-coupled chain, upstream information is worth 12.4% of closed-loop cost, and the mechanism that captures it is an affine delay line: what matters is that every node receives its warning with the *same* anticipation, which a proportional delay cannot provide and a constant term can. In a fleet of loops sharing one embedded controller, governing the horizon is worth 5–6% at equal spent computation, and the governor that achieves it is simply a detector of certificate violation behind a slow leak: two effective parameters. In both cases everything more sophisticated, including the field we were defending, is statistically no better or significantly worse.

The second thing that survived is a set of structural statements that say, in advance, why the sophisticated alternatives could not have won (Section IV). They are elementary: a homogeneity argument, the observation that a linear time-invariant operator is a fixed reshaping, a DC-gain computation. Each one, once stated with its hypotheses complete, retired an entire family of test benches or mechanisms, several of them ours.

The third is methodological, and we believe it travels furthest (Section VIII). Every arena we built contained at least one defect that inverted or invalidated its first conclusion. Five of those defects form a pattern worth naming: an acausal time shift hiding in a tuning grid, an acausal normalization hiding in a scale factor, a broken channel hiding under a plausible forcing term, and two accounting slips, one of which inflated the result it touched while the other, more insidiously, *deflated* it. Nearly all were found by automated adversarial review rather than by us; one we found ourselves, late. Each is now an executable assertion, and the tests that catch them (a pulse probe, a truncation probe, a DC-gain probe, spent-computation accounting, boundary-optimum rejection) are cheap enough to run inside a tuning loop.

A sign error we found in the multiparametric machinery of a widely cited multiobjective MPC formulation [1] is stated and proved in Section III, with its consequences for that formulation's weight-selection subproblems; Section II uses only the piece that matters for implementation. The authors of [1] were contacted about the observation prior to this submission; no response had been received at the time of writing.

Relation to existing work

Feedforward compensation of a *known* transport delay is classical, and it is standard practice in the domains our transport arena abstracts: district heating networks [2] and distributed solar collector fields [3], [4]. We do not claim novelty for the idea that delayed upstream information helps. The contribution is narrower and, we think, more transferable: when the plant model is kept nominal and the only channel available is the meta-parameter, the *structure* of the delay family decides everything — the proportional family that a per-hop design naturally produces is structurally unable to deliver constant anticipation, an affine family is, and nothing richer (second-order shaping, a wave field on the graph) improves on it. We found this because our own "unreachable oracle" turned out to be implementable, which is itself a methodological point.

Distributed and hierarchical MPC architectures are surveyed in [5], [6], with plug-and-play variants in [7] and canal networks in [8]. Abstracting a chain of subsystems by a PDE and reasoning about propagation on it appears in vehicular platooning [9], and wave-equation-based formation control exists [10]; our negative results on the graph field do not touch those uses, which couple the *physical* states rather than the meta-parameters. On horizon adaptation, variable and adaptive horizons with guarantees are treated in [11], [12], and a robust self-triggered scheme in which the horizon adapts as part of the triggering mechanism appears in [13]. We contribute no new guarantee; we characterize empirically *which* governor decodes the suboptimality signal, and we connect the winner to the certificate itself.

Contributions

1) **The affine-delay result** (Section V): in a transport-coupled storage chain, constant-anticipation delivery through the meta-parameter channel recovers 90.9% of a clairvoyant bound; the proportional-delay family, a matched shaped filter, and a wave field are structurally or statistically inferior, with the failure of the proportional family explained.
2) **The certificate-detector result** (Section VI): a horizon governor that maps certificate violation ($\alpha_t \leq 0$) to maximal horizon and decays otherwise sits below the fixed-horizon frontier in a single loop and beats the best fixed horizon by 5–6% in a fleet at equal spent computation, robustly to disturbance correlation; with a clean decomposition of temporal versus allocation gains.
3) **Structural limits** (Section IV) with hypotheses stated in full and their necessity demonstrated by counterexample.
4) **An audit methodology** (Section VIII): five real leaks, the conclusion each inverted, and the executable test that now guards each; all distributed as a regression suite.
5) **A sign correction** (Section III): the value function of the weighted multiparametric program at the core of the multiobjective formulation of [1] is *concave*, not convex, in the weights; their weight-selection LP is a safe inner approximation rather than an equivalent program, their quadratic-branch online problem is not a convex program, and the exact remedy — enumeration or gridding — is what Section II implements.

II. Problem Setting

Consider the discrete-time linear system

$$x_{k+1} = Ax_k + Bu_k + w_k, \qquad x_k \in \mathbb{R}^n,\ u_k \in \mathcal{U}, \qquad (1)$$

with $\mathcal{U} = \{u : \|u\|_\infty \leq u_{\max}\}$ and w_k an unmeasured disturbance. Throughout, the MPC is *nominal*: it predicts with $x^+ = Ax + Bu$ and no disturbance model, which is both standard and the honest setting for our question — with a disturbance model the controller would compensate by feedforward and there would be nothing left for a meta-parameter to arbitrate.

A. Weight-governed multiobjective MPC

Two stage costs $\ell_i(x, u) = x^\top Q_i x + u^\top R_i u$ compete, scalarized on the simplex with weight $(1-\alpha, \alpha)$, $\alpha \in [0, 1]$. With terminal weights P_i from the discrete Lyapunov equation for a common terminal gain K_f and an invariant terminal set from the maximal output admissible set construction [14], the controller solves

$$V^\star(x, \alpha) = \min_{U \in \mathcal{U}^N \cap \mathcal{X}_N(x)} (1-\alpha) J_1(x, U) + \alpha J_2(x, U), \quad (2)$$

and the applied weight is restricted by the contractive constraint inherited from the weighted-sum multiobjective formulation of [1], $V^\star(x, \alpha) \leq J_a$ with J_a the cost of the shifted sequence under the previous weight. One parametric fact matters for implementation: for fixed x the map $\alpha \mapsto V^\star(x, \alpha)$ is a pointwise minimum of functions affine in α, hence *concave* [15, Sec. 5.6.1, Ex. 3.9], so the admissible weight set $\mathcal{A}(x, J_a) = \{\alpha : V^\star(x, \alpha) \leq J_a\}$ is in general non-convex — in our instances it was non-convex at roughly one fifth of the sampled contraction levels — and must be gridded or enumerated rather than projected onto. The sign, its origin, and its consequences for the original formulation are the subject of Section III.

B. Horizon-governed MPC and the suboptimality index

For the single-objective case we use the relaxed dynamic programming index of Grüne and Pannek [16]–[18]. With V_N the optimal value of the N-step problem *without* terminal ingredients and ℓ_t the incurred stage cost,

$$\alpha_t = \frac{V_N(x_t) - V_N(x_{t+1})}{\ell_t}, \tag{3}$$

and a *uniform* $\alpha_t \geq \alpha > 0$ certifies $J_\infty \leq V_N/\alpha$ [17]. Evaluated along a closed loop, (3) is an a posteriori diagnostic, not a per-step certificate: it can be negative, and the telescoping identity it satisfies is accounting rather than a guarantee. The event $\alpha_t \leq 0$ — the certificate breaking — turns out to be the one feature of this signal a governor should react to (Section VI).

Because (3) is a ratio it is uninformative when $\ell_t \approx 0$, and gating on ℓ_t alone silently annuls exactly the sample created by a disturbance kick, whose pre-kick stage cost is essentially zero; our first implementation did precisely that and produced governor traces such as $N = [2, 2, 16, 2, 16, \dots]$. A jump in V_N betrays an event even when ℓ_t vanishes, so the informativeness gate must be bilateral.

C. The graph field, i.e. the elaborate governor under test

The distributed governor we set out to defend assigns each subsystem i a field value h_i that becomes its meta-parameter request. With L the graph Laplacian and A the adjacency matrix,

$$K = \omega_0^2 I + c^2 L, \quad C = 2\zeta\omega_0 I + DL, \quad G = bA + \beta A^{\circ 3}, \tag{4}$$

a wave branch $\ddot{h} = -(C+G)\dot{h} - Kh + f(h_d)$ integrated by Störmer–Verlet, and a diffusion branch $\gamma\dot{h} + (K+G)h = f(h_d)$ integrated by an IMEX scheme. The two branches share K, hence the steady-state profile, so wave-versus-diffusion is a mechanism null by construction. The choice of forcing $f(\cdot)$ looks like a detail; Proposition 3 shows it decides whether the field can transport anything sustained at all.

III. A Sign Correction to the Multiparametric Machinery of a Weighted Multiobjective Formulation

This section is self-contained and concerns only the multi-objective formulation our weight governor builds on. Lemma 4 of [1] considers the multiparametric linear program — their problem (13) — with weights μ entering the cost and the state x entering the right-hand side, and asserts that the value function is "convex and piecewise affine w.r.t. μ for any given x." Piecewise affinity is correct, and so is the statement in x; the convexity claim in μ has the wrong sign.

Lemma 1. *Let $V^*(\mu, x) = \min_z\{(c_0 + C_\mu^\top \mu)^\top z : \ Gz \leq b + Sx\}$ be the value function of problem (13) of [1]. For each fixed x with nonempty feasible set and attained minimum, $\mu \mapsto V^*(\mu, x)$ is* concave *and piecewise affine on its domain.*

Proof. For each fixed feasible z the map $\mu \mapsto (c_0 + C_\mu^\top \mu)^\top z$ is affine, and the feasible set does not depend on μ; hence $V^*(\cdot, x)$ is a pointwise minimum of affine functions of μ, therefore concave, and — the minimum being attained at one of finitely many vertices, or the problem being unbounded — a minimum of finitely many of them where finite, therefore piecewise affine on its domain. □

This is the classical sensitivity dichotomy of parametric programming: under minimization the optimal value is convex in right-hand-side perturbations — the side treated by classical multiparametric linear programming [19] — and concave in parameters entering the objective linearly [15, Sec. 5.6.1, Ex. 3.9]; both signs invert under maximization. The slip traces to the companion conference paper [20, p. 2406], which establishes the properties in x and adds that "by duality, the same properties hold with respect to μ": duality does exchange cost and right-hand-side parameters, but it also turns the primal minimum into a dual maximum, and that is the step that flips the curvature. We also confirmed the sign numerically, on two families. Over 300 random chord tests on quadratic instances with this parametric structure, the concavity inequality in μ held in every case and the convexity inequality failed in every case. Over 1,200 chord tests on randomly generated instances of their problem (13) itself, strict concavity appeared in 791 cases and strict convexity in none, the remaining chords falling on affine pieces where both inequalities hold with equality; the same test in x confirmed convexity — exactly the corrected statement.

Consequence for their Theorem 6. The proof invokes [21] for *convex* piecewise affine functions to evaluate $V^*(\mu, x)$ as the *maximum* of its affine pieces, and the weight-selection LP — their (17) — imposes every piece $\leq J_a$. By Lemma 1 the value function is the *minimum* of those pieces, so (17) describes a convex *inner approximation* of the true admissible weight set $\mathcal{A}(x, J_a) = \{\mu : V^*(\mu, x) \leq J_a\}$, which is a sublevel set of a concave function and in general non-convex. One consequence is reassuring and one is not: every weight accepted by (17) satisfies the contractive constraint, so all closed-loop stability guarantees *built on that constraint* are preserved — the piecewise affine branch is safe as published; but the claimed equivalence fails, (17) can be infeasible while admissible weights exist, and its optimizer need not be the admissible weight closest to the desired one. In our instances the admissible set was non-convex at roughly one fifth of the sampled contraction levels, so the gap is not a measure-zero curiosity.

Consequence for their Theorem 7. For the branch with a quadratic objective, joint convexity of the value function in (μ, x) is asserted "by the results of Mangasarian and Rosen," and the online problem — their (21) — is concluded to be a convex program. The cited result [22] concerns perturbations of the constraint right-hand side; in their problem (20), however, μ enters the objective linearly while the constraints do not depend on it, so the argument of Lemma 1 applies verbatim: the value function is concave in μ, joint convexity cannot hold except degenerately, and (21) is not a convex program. Unlike the piecewise affine case, no inner-approximation safety net arises here, so any weight returned by a solver should be certified a posteriori by evaluating the value function directly — which is cheap.

Remedy, and what this paper does. For the low weight dimensions typical of scalarized multiobjective problems [23], [24], the admissible set is handled exactly by enumeration over the affine pieces or by gridding the simplex and testing $V^*(\mu, x) \le J_a$ pointwise; projections, gradient steps on the weight, and any construction presuming convexity of $\mathcal{A}(x, J_a)$ can silently leave the admissible set. That is what Section II already implements, and it is the only point where the correction touches the rest of this paper. The contribution of [1] — closed-loop guarantees through a contractive constraint on a weighted value function — is not in question: the correction concerns the exactness claims attached to the weight-selection subproblems.

IV. Structural Limits

The statements in this section are elementary. We give them anyway, with hypotheses complete, because each one retired a family of benches or mechanisms, and because in two cases the hypothesis we originally omitted turned out to be the load-bearing one.

A. Amplitude-scale invariance

Proposition 1. *Consider* (1) *with quadratic stage costs, zero initial state $x_0 = 0$, and every constraint — input, state, terminal — inactive along the whole trajectory, for every meta-parameter value swept and every scale considered. Fix the meta-parameter (weight α or horizon N). If the disturbance is scaled $w \mapsto \lambda w$, $\lambda > 0$, the optimal trajectories scale by λ, the cost by λ^2, and therefore the entire normalized cost curve $J(\cdot)/\lambda^2$ — hence also its set of minimizers — is independent of λ.*

Proof. With no active constraint the closed loop is linear in the exogenous input for each fixed meta-parameter, and with $x_0 = 0$ there is no free response to break homogeneity; a quadratic cost then scales by λ^2, and a positive common factor moves nothing. □

Numerically, with both optima interior to their grids ($\alpha^\star = 0.20$, $N^\star = 32$), the normalized cost curves for $\lambda \in \{0.5, 1, 2, 4, 8\}$ coincide to relative dispersion 5.4×10^{-14} (weight) and 1.2×10^{-16} (horizon) (Fig. 1). The hypotheses are not decorative: with $x_0 = (0, -4)$ on the same strictly linear plant the optimal horizon moves with λ and the normalized cost disperses by order one. And the statement is about the *curve*, deliberately: when the minimum is flat, the literal argmin can hop between points whose costs differ by 10^{-16}, so comparing argmins is the wrong test — our own audit assertion failed for exactly that reason until we rewrote it to compare curves and near-optimal sets.

What follows is bounded, and our first draft over-claimed it. Homogeneity forbids the optimal meta-parameter from depending on the *absolute amplitude* of the disturbance; it does not forbid dependence on the shape or phase of a normalized envelope, which are scale-invariant. So a bench is empty exactly when its demand signal is *postulated as a function of absolute amplitude* — as was the fatigue-derived schedule $\alpha_d = 1/(1 + \kappa E^2)$ of our narrow-band arena, which duly lost in all eight configurations (Fig. 6).

B. The graph field is an LTI filter bank, with a scope warning

Proposition 2. *With $b = \beta = 0$ in* (4)*, the map from the injected signal h_d to the node values h, in either branch, is linear and time invariant: each node's response is the convolution of the source with a fixed impulse response. (Verified to 1.1×10^{-15} by superposition, exactly by time invariance, and to 4.4×10^{-16} by convolution reconstruction.)*

Whatever the field contributes over the signal entering at the source is therefore a fixed linear reshaping — delay plus spreading. Two honest caveats, both of which we got wrong before getting them right. First, the field belongs to the class of LTI banks of order $2M$; a rival parametrizing that whole class would contain it, but the seven-parameter shaped delay we use in Section V does *not*: measured on interior hops of an eight-node chain, the field's kernels broaden with hop count by a factor of 1.2 to 1.8 over four hops depending on the measure (absolute-value dispersion, energy dispersion, half-amplitude width), at the operating point as well as at nominal parameters. The field's defeat there is an empirical finding, not a corollary. Second, the deployed governor is *not* LTI — between h and the applied meta-parameter sit a saturation, a projection onto $\mathcal{A}$, and a quantizer — and the reach law below depends on the quantizer.

C. The forcing decides whether anything crosses

Proposition 3. *Take the forcing $f(h_d) = K h_d$, so the stiffness term reads $-K(h - h_d)$ and each node relaxes to its own target. Then the equilibrium is $h = h_d$ exactly, $H(0) = K^{-1}K = I$, and the cross-node DC gain is identically zero: a sustained alert injected at one node reaches no other node, under any tuning. With $f(h_d) = \omega_0^2 h_d$ the equilibrium is $\omega_0^2 K^{-1} h_d$, whose off-diagonal entries are nonzero, and with $G = 0$ both branches share the same $H(0)$, preserving the mechanism null.*

A unit step at node 1 of a six-node chain settles to $[1, 0, 0, 0, 0, 0]$ to machine precision under the first forcing (Fig. 2); only edges cross, as transients. Our transport experiment originally asked the field to deliver a sustained window under exactly this forcing — the one thing its structure forbids — and the field's tuned gains pinned themselves to the top of every grid trying to compensate, which was the observable symptom.

Observation 1 (Quantization-limited reach)**.** If the closed-loop transported amplitude decays geometrically with hop count, $a_n = a_0 \rho^n$, and the meta-parameter moves only when that amplitude exceeds one grid step Δ, the reach is $n_{\max} = \lfloor \log \Delta / \log \rho \rfloor$: logarithmic in the resolution of the meta-parameter.

Measured on an eight-node chain with $\rho = 0.358$ fitted on interior nodes, the step law matches at three of four resolutions and overpredicts by one hop at the finest, where the disputed amplitude sits within the resolution of the measurement (Fig. 3). Two cautions we needed: the reach is an integer, so plotting the continuous expression against integer measurements makes a correct law look wrong; and the terminal

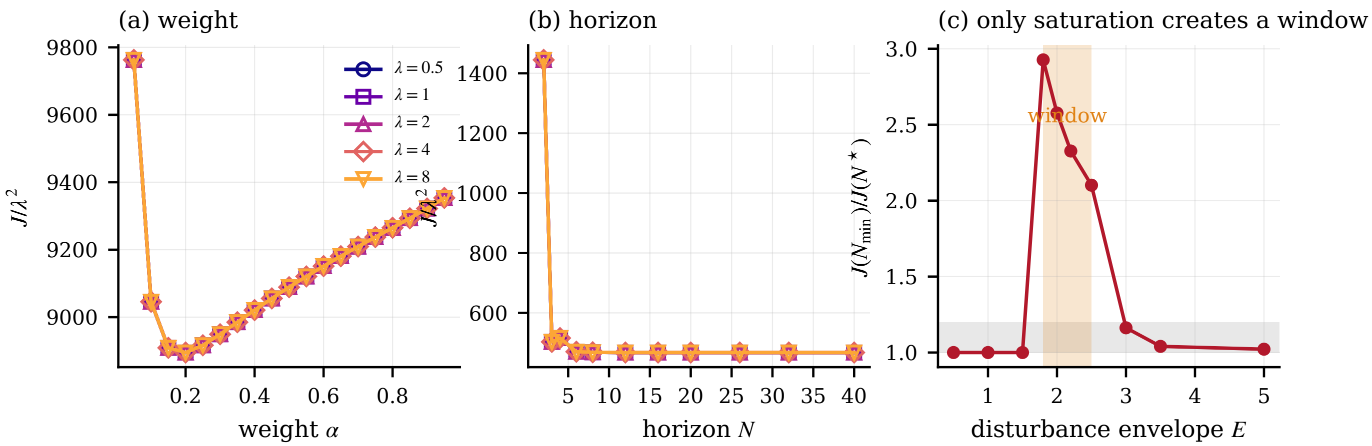

Fig. 1. Amplitude-scale invariance and its limit. (a,b) Normalized cost J/λ^2 against weight and horizon for five disturbance scales, both optima interior ($\alpha^\star = 0.20$, $N^\star = 32$): the curves coincide. (c) With a saturating actuator a level-dependent optimum does appear, but only in a narrow window of the disturbance envelope.

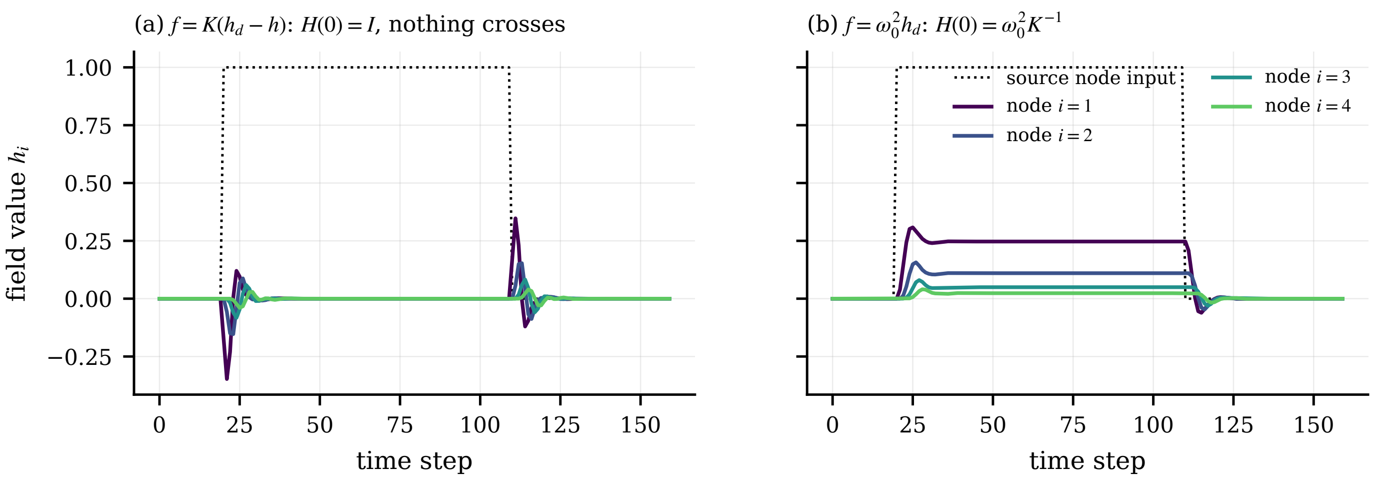

Fig. 2. Proposition 3. (a) Under the homeostatic forcing the DC transfer is the identity: a sustained input at the source produces nothing at any other node. (b) Under the source forcing the sustained component crosses, identically in both branches when $G = 0$.

node of a chain has degree one and retains signal, so fitting ρ through it biases the estimate — an artifact that, we admit, also contaminated one of our own intermediate re-measurements before we excluded terminal nodes throughout. The half-step threshold that round-to-nearest quantization suggests fits none of the four resolutions; the amplitude we report is a dispersion along the trace, not a peak, and the empirical criterion is the full step.

V. Result I: Constant Anticipation, Not Proportional Delay

The one use of a graph field that the structural limits leave open is transport: a disturbance that travels physically downstream, so that the upstream neighbour holds information about the downstream node's future which no local horizon can obtain. Node i experiences the source disturbance delayed by $i\tau$ steps ($\tau = 6$ in our chain of four nodes).

A. Three viability filters

Three checks preceded any comparison, each able to kill the arena. Without a storage state, the lead sweep is flat: a nominal regulator at rest applies $u^\star = 0$ for every meta-parameter value, so advance warning has no channel to act through — unless the governor itself has lag, which is why we state this as a filter rather than a theorem. With storage — two objectives at different setpoints, so the weight selects the buffer level and anticipating means charging early — anticipation buys $+4.0\%$ over the ideal reactive limit ($t = +5.80$, $p = 2.6 \times 10^{-4}$, nine of ten seeds, tuning and evaluation seeds disjoint). The useful lead is short, two to four steps across the physical range, and grows with actuator sluggishness as the mechanism predicts.

B. The discovery: our unreachable oracle was implementable

We compared delivery mechanisms for the source measurement: an ideal local detector (learns of the event the instant it arrives — stronger than any realizable detector), a per-hop delay line $k_i = \delta i$, the wave field, a shaped delay (per-hop delay plus a second-order stage, matched to the field's seven parameters), and, as a reference ceiling, a "clairvoyant oracle" that opens a gate a fixed lead before each arrival.

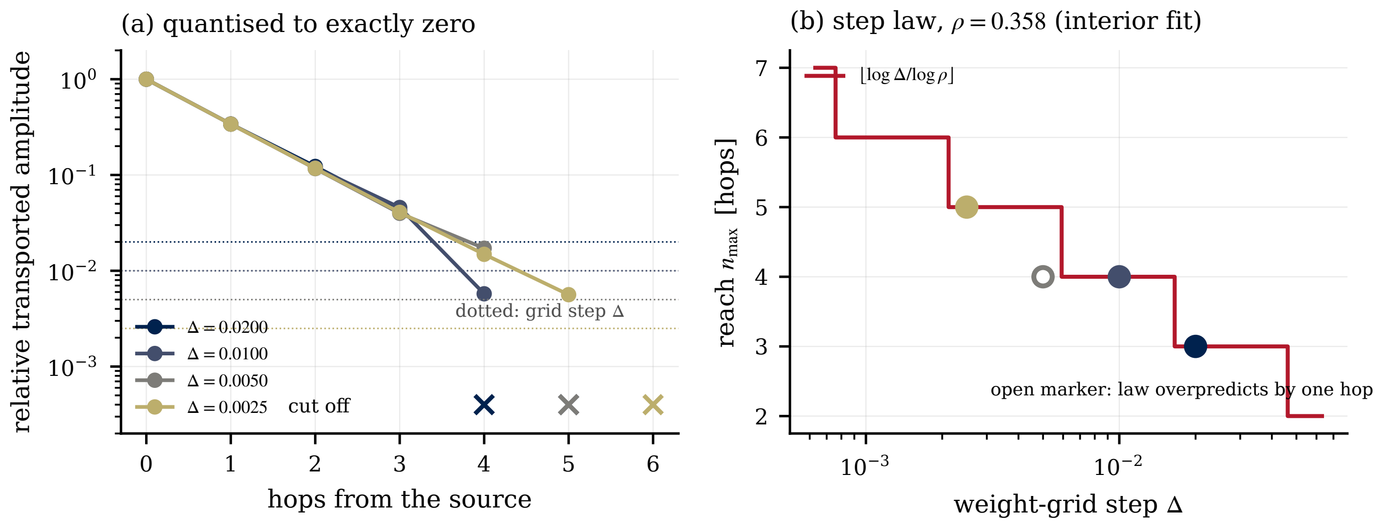

Fig. 3. Observation 1. (a) Transported amplitude against hop count for four grid resolutions; crosses mark nodes quantized to exactly zero, dotted lines the grid step. (b) Measured reach against grid step, with the step law and $\rho = 0.358$ fitted on interior nodes; the open marker is the one resolution where the law overpredicts by a hop.

Getting this comparison honest took three corrections, described in Section VIII; here is the one that produced the result. After the causality guard was in place, the tuned oracle — supposedly an unreachable bound — opened its gates at $t = 200, 206, 212$ for arrivals at $206, 212, 218$: never before the source measured. It was causal, and it was reproduced *exactly* (difference 0.0) by a delay line

$$k_i = \max\{0,\ \tau i - \ell\}, \tag{5}$$

affine in the hop index with a constant anticipation term ℓ. The competition had excluded the best member of its own family. The proportional family $k_i = \delta i$ cannot express (5): with $\delta = \tau$ the anticipation is zero everywhere, and with $\delta < \tau$ it grows with distance, so far nodes are warned too early and near nodes too late. What the physics supplies is anticipation *growing* with distance ($i\tau$); what the loop needs is anticipation *constant* across nodes (saturated, for near nodes, at the maximum physically available); the constant term is the conversion between the two.

TABLE I
TRANSPORT CHAIN, 30 EVALUATION SEEDS. "MDE" IS THE MINIMUM DETECTABLE EFFECT AT POWER 0.8 FOR THE PAIRED CONTRAST; CONTRASTS ARE HOLM-CORRECTED [25]. THE AFFINE DELAY AGAINST THE PROPORTIONAL FAMILY'S *best known* TUNING (FROM A DENSER GRID, ARCHIVED) STILL GIVES $t = +5.40$, SO THE DECISIVE CONTRAST DOES NOT DEPEND ON THE RIVAL'S VERSION.

mechanism	cost (s.e.m.)	vs. local	% ceiling
acausal ceiling	48 944 (2828)	+13.7	100.0
affine + shaping (7 par.)	49 455 (2175)	+12.8	93.4
affine delay (4 par.)	**49 650** (2370)	**+12.4**	**90.9**
wave field (7 par.)	51 054 (2083)	+9.9	72.8
proportional delay (3 par.)	51 711 (2479)	+8.8	64.3
ideal local detection	56 690 (1301)	0.0	0.0
paired contrasts, Holm; δ, t, p, MDE as % of cost			
affine vs. proportional	+2060, t=+17.87, p<10^{-4}*, 0.6%		
affine vs. wave field	+1403, t=+4.64, p=0.0002*, 1.5%		
shaping the affine	−195, t=−0.97, p=0.34, 1.0%		
affine vs. local	+7040, t=+6.29, p<10^{-4}*, 5.5%		
ceiling vs. affine	+707, t=+1.49, p=0.29, 2.3%		

C. The result

Table I and Fig. 4 give the final comparison: thirty evaluation seeds disjoint from the three tuning seeds, causality guard active inside the tuning loop, causal (parameter-only) normalization, equal tuning-grid densities for all mechanisms, and every tuned optimum interior to its grid except at physical simplex limits. The genuinely acausal ceiling uses per-node anticipation beyond what the source can supply and is labelled as a bound, not a mechanism.

Three readings. Having upstream information is worth 12.4% against the best possible local detector, and the study can resolve effects an order of magnitude smaller than that. The *structure* of delivery decides: affine beats proportional at $t = +17.87$ under equal tuning densities, and at $t = +5.40$ even against the proportional family's best tuning from a denser grid. And nothing richer helps: adding the second-order shaping stage changes nothing resolvable ($t = -0.97$ against an MDE of 1.0%), the wave field is significantly worse than the plain affine line ($t = +4.64$), and the affine line is statistically at the ceiling ($p = 0.29$, MDE 2.3% — an equivalence bound, not a proof of optimality). We state the one asymmetry that survives: the field's kernel does differ across hops (Proposition 2 discussion), so its defeat is empirical, bounded by these MDEs, not structural.

VI. RESULT II: A CERTIFICATE DETECTOR GOVERNING THE HORIZON

A. Single loop: the frontier, and who sits below it

We first close the loop $\alpha_t \to$ governor $\to N_t$ on a single plant, with pre-registered hypotheses, ten seeds, and computation accounted honestly: every QP is charged, including those

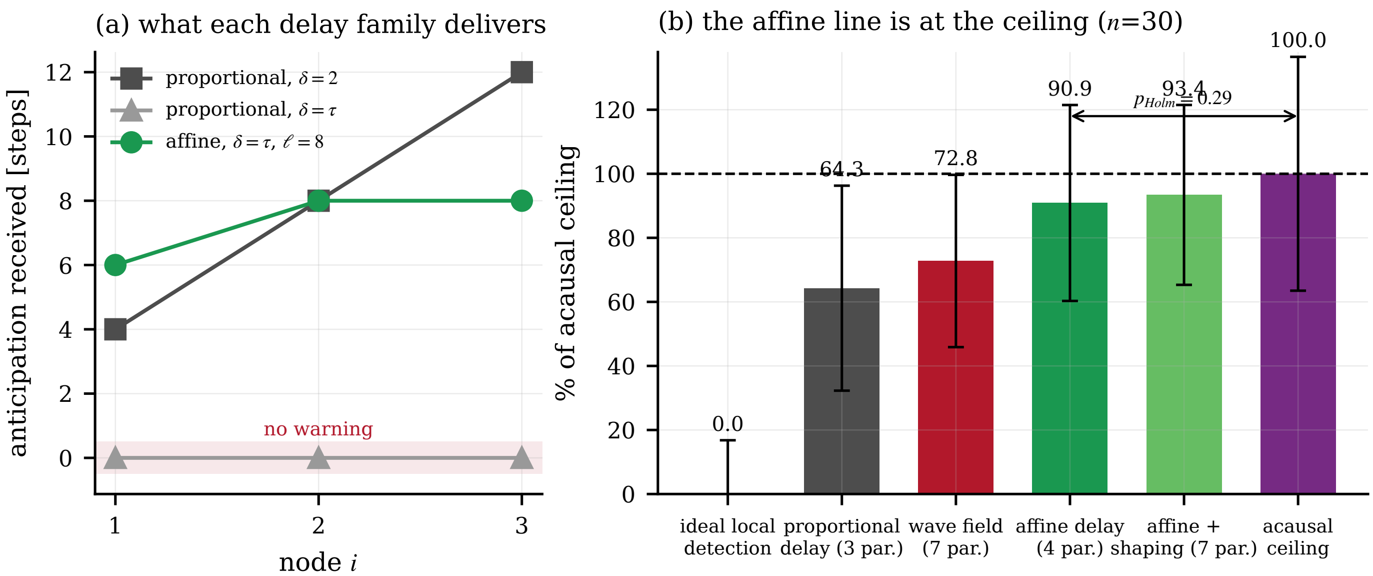

Fig. 4. (a) Anticipation received by each node under the two delay families: the proportional family either wastes the near nodes' information or floods the far nodes early; the affine family delivers constant anticipation, saturated at what is physically available. (b) Final comparison, thirty seeds, error bars one s.e.m.: the affine delay is statistically at the acausal ceiling, and neither shaping nor the wave field improves on it.

TABLE II

Horizon governance, out-of-envelope scenario, ten seeds. Divergent runs are excluded pairwise from contrasts and reported as counts; the $N = 2$ row's computation and cost are means over a run set with half the seeds dead, and are shown for completeness rather than comparison.

governor	comput.	cost	corr	div.
fixed $N = 2$	610	29653	—	5
fixed $N = 4$	1600	12561	—	0
fixed $N = 6$	2400	12124	—	0
fixed $N = 8$	3200	12104	—	0
fixed $N = 16$	6400	12103	—	0
leaky integrator	**1775**	**12211**	0.935	0
threshold	3051	12103	0.858	0
2nd-order field (ζ=0.5)	3133	17862	0.270	2
memoryless map	3666	27620	0.367	5

used to measure α_t, with the instrument solve reused as the next control solve when the horizon does not change. Table II reports the out-of-envelope scenario (larger kicks plus model mismatch) against the frontier traced by fixed horizons at equal computation.

Interpolating the frontier at the leaky integrator's 1775 units gives roughly 12 470; the governor achieves 12 211, about 2% below the frontier, without knowing the correct horizon in advance. The second-order field rings — it drops the horizon mid-recovery, breaks the certificate again and re-excites itself; critical damping repairs its assignment correlation in the large-kick scenario (-0.42 to $+0.43$, $t = +7.26$), which identifies ringing as the mechanism, but a critically damped second-order filter is a first-order filter with extra lag and does not beat one. On assignment correlation the leaky integrator is best in two of the four scenarios, including both that combine kicks with mismatch. What broke out-of-envelope was not short *fixed* horizons at moderate N — replanning compensates from $N = 4$ up, though $N = 2$ does diverge on half the seeds — but short-*memory adaptive* governors, which under-assign right after a kick. The risk is aggressive downward modulation, not a modest fixed horizon.

The demand signal α_t generates is a pulse train; the natural decoder of pulse durations is a leaky integrator, not a resonator. Pushing the governor's reference toward zero turned out to improve it monotonically, with a plateau at the limit: the optimal governor is a *detector of certificate violation* — request $N_{\max}$ when $\alpha_t \leq 0$, request $N_{\min}$ otherwise — behind a slow leak ($\tau \approx 30$ samples). Two effective parameters, and the event it reacts to is the one event in the signal with a meaning in the theory [16].

B. A fleet sharing one embedded controller

A factor-two saving in computation matters most where predictive control is most constrained: one embedded controller serving many loops, the sampling period fixing a hard computation budget per step. We model $M = 8$ thermal collector loops (fluid state coupled to a slower absorber, $\tau_f = 2\,\text{min}$, $\tau_m = 5\,\text{min}$, $dt = 30\,\text{s}$), with cloud-passage disturbances and a knob ρ setting the fraction that sweeps all loops at once. The penalized variable is the *absorber* temperature, which the flow reaches only through the fluid; without that delay between action and effect the bench is empty — penalizing the directly actuated state gives $J(N{=}2)/J(N{=}16) = 1.01$ and nothing to govern, against 1.46 here.

Two accounting disciplines decide whether this comparison means anything, and we broke both before fixing them (Section VIII). The budget must bind on computation actually *spent* — including the certificate measurement — not on horizons granted; and the governor must be *tuned* under the same accounting it is evaluated under. With both in place, the rival is the fixed-horizon frontier interpolated, per seed, at the

TABLE III
FLEET OF $M = 8$ LOOPS, EIGHT EVALUATION SEEDS DISJOINT FROM THE THREE TUNING SEEDS, CERTIFICATE-DETECTOR GOVERNOR ($\tau = 30$, REFERENCE AT THE DETECTOR LIMIT; BOTH INTERIOR, THE REFERENCE ON A VERIFIED PLATEAU). COST AGAINST THE FIXED-HORIZON FRONTIER AT EQUAL SPENT COMPUTATION, PAIRED PER SEED. THE $B = 16$ ROW IS DEGENERATE: EVERY LOOP IS PINNED AT $N_{\min}$ AND THE $+0.03\%$ IS SOLVER NOISE WITH PAIRED VARIANCE NEAR ZERO.

$B/(MN_{\max})$	spent	governed	frontier	change
independent disturbances ($\rho = 0$)				
0.19	26.0	2814	2993	$\mathbf{-6.0\%}$ *
0.25	35.2	2567	2732	$\mathbf{-6.0\%}$ *
0.31	41.1	2470	2598	-4.9% *
0.38	42.9	2443	2568	-4.9% *
≥ 0.50	43.1	2440	2564	-4.9% *
common front ($\rho = 1$)				
0.19	23.8	4589	4599	-0.2% *
0.25	29.1	4208	4343	-3.1% *
0.31	33.5	3912	4149	$\mathbf{-5.7\%}$ *
0.38	37.3	3750	4003	$\mathbf{-6.3\%}$ *
0.50	43.1	3645	3829	-4.8% *
≥ 0.75	48.2	3638	3716	-2.1% *

governor's real computation; interpolation between adjacent integer horizons is achievable by alternating horizons between steps, so it concedes nothing.

Table III and Fig. 5: the governor beats the best fixed horizon at equal spent computation in seven of eight budgets, by up to 6.0% with independent disturbances ($p < 10^{-5}$ throughout) and — this surprised us — by up to 6.3% under a perfectly correlated front ($p < 0.03$ throughout). The robustness to correlation has a clean explanation, which the decomposition in Fig. 5(b) isolates by comparing demand-proportional against equal-split allocation under the same governor: the *temporal* mechanism (maximal horizon only while a certificate is broken) pays regardless of correlation, because it lives inside each loop; the *allocation* mechanism (moving budget toward loops in trouble) pays up to $+8.6\%$ under scarcity when disturbances are independent ($+5.5\%$ or more at the three tightest budgets, falling to $+1.7\%$ as the budget loosens), and is neutral to marginally negative under a common front, when every loop's demand is the same and there is nothing to multiplex. The multiplexing intuition is right, but it lives in the allocation component only.

VII. THE ARENA MAP

Table IV summarizes the six arenas. The one the field won is the one whose demand was an exogenous priority signal with no physical origin, and we do not count it as evidence.

There is a tension worth stating plainly, because it bounds future attempts. For modulation to pay one must cross a nonlinearity; crossing one makes the demand eventual, hence broadband, which is the regime where a leak beats a resonator. For inertia to pay one wants narrow-band demand, whose canonical manufacture is amplitude modulation, which Proposition 1 strips of absolute-amplitude content. The two conditions an inertial governor needs sit awkwardly together, and in six arenas they never met.

TABLE IV
THE SIX ARENAS. "DEMAND" IS THE SIGNAL THE GOVERNOR IS ASKED TO TRACK.

arena	demand	outcome
synthetic weights	exogenous priority, no physics	field wins out of distribution; no physical origin, not counted
greenhouse price	smooth, clean	modulation loses; the horizon already arbitrates the price
graph, shared price	global signal	no gain: sharing what is already common
α_N horizon	pulse train	pays; certificate detector + leak wins, the field rings
narrow-band vibration	postulated from absolute amplitude	empty by Proposition 1: loses $8/8$
transport, storage	travelling event	pays 12.4%; affine delay wins, field and shaping add nothing

VIII. FIVE LEAKS, AND THE TESTS THAT NOW GUARD THEM

Every arena above was first decided by a defect rather than by its mechanisms. We document the five that form a pattern. None of them is exotic; each is the kind of line a careful person writes without noticing, and each has a mechanical guard cheap enough to run inside a tuning loop. Table V lists the resulting suite.

a) Leak 1: a broken channel (DC cross-gain): The field was first run with the homeostatic forcing, whose cross-node DC gain is identically zero (Proposition 3); the experiment asked it to deliver a sustained window, the one signal it cannot carry, and the field lost to a trivial delay line. The symptom was tuned gains pinned to grid boundaries. *Guard:* probe the channel with a sustained step and measure what arrives, before comparing anything (assertion A18).

b) Leak 2: an acausal time shift: A free time-shift in the signal-conditioning stage is a natural knob to include in a tuning grid; if the grid contains negative values, it silently hands the mechanism information from the future. Both leading mechanisms took the option — their gates opened up to eight steps before the source measured the event — and the only fair player was the simplest rival. The headline contrast of that version ($t = +5.45$) dissolved to $t = +0.05$ under the guard. *Guard:* a single-pulse probe inside the tuning loop; reject any configuration whose output moves before the input does.

c) Leak 3: an acausal normalization: The map from field value to weight was normalized by the trace's min and max — computed over the *whole* trace, so the scale at time t depended on the future. Truncating the trace after the event changed the gate *before* the event by 0.42. Only the two seven-parameter mechanisms used the normalization; the delay line did not. *Guard:* the truncation probe — run the mechanism on the full and on a truncated input; on the shared prefix the outputs must agree exactly. This test requires no hypothesis about where the leak is, which is how it caught one we did not suspect. The fix

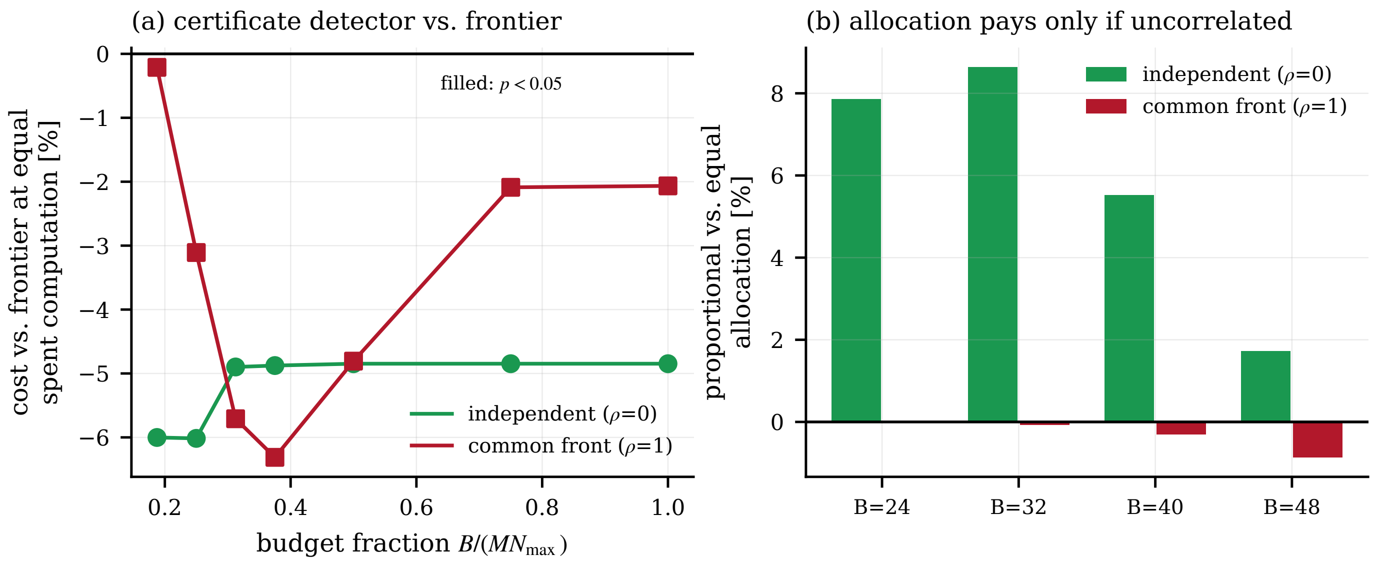

Fig. 5. (a) Fleet cost against the fixed-horizon frontier at equal *spent* computation, for independent and perfectly correlated disturbances: the certificate-detector governor wins across budgets in both. (b) Decomposition: demand-proportional versus equal-split allocation under the same governor. Allocation pays only when disturbances are independent; the temporal mechanism accounts for the rest.

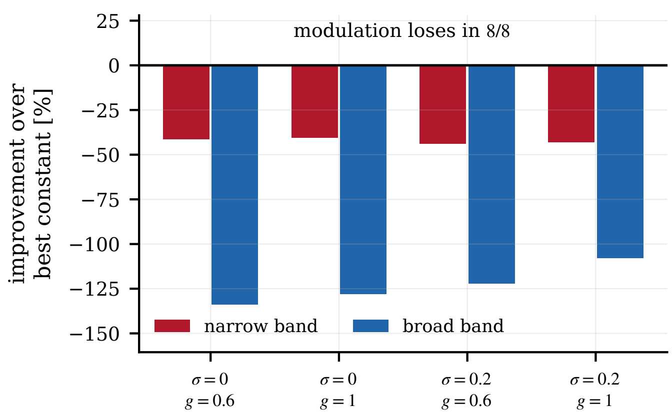

Fig. 6. The narrow-band arena, built expressly to favour the field: resonant structure, beat disturbance, fatigue cost, and a schedule $\alpha_d = 1/(1 + \kappa E^2)$ *derived* from that cost. It loses in all eight configurations against a per-condition re-optimized constant weight — as Proposition 1 predicts for any schedule postulated from absolute amplitude.

(normalizing by the DC gain, a pure function of parameters) also removed a free knob.

d) Leak 4: budget granted versus computation spent: The fleet's budget bound the horizons *granted*, while the governor also spent computation measuring its certificate; it consumed 18–26% more than the fixed rival at the same nominal budget, and "won" by overspending. *Guard:* account computation where it is spent and compare on the (spent, cost) plane. The single-loop experiment already did this; the fleet inherited the budget from the wrong side of the ledger.

e) Leak 5: tuning under one metric, evaluating under another: After fixing Leak 4 we re-tuned the governor by raw cost at fixed nominal budget — a criterion that still rewards overspending — and concluded the fleet result had deflated to nothing. Tuned under the evaluation metric itself (cost against the frontier at spent computation), the governor wins by 5–6%. This leak is the mirror image of the others: it *deflated* the result it touched. Leaks do not consistently favour the pet mechanism; they favour whatever the criterion mismatch happens to favour, which is why the guard must be mechanical. *Guard:* the tuning objective must be the evaluation metric, computed the same way.

f) Three disciplines, used throughout: Reject any tuned optimum on the boundary of its grid, with the exception of physical limits (a weight saturating at the simplex boundary is not a grid artifact) — applied to ourselves twice, including once to the evidence for Proposition 1. Report, with every null contrast, the minimum detectable effect: "no difference" in Table I means "bounded by the stated MDE," never "proved zero." And treat an "unreachable" bound that a guard fails to disqualify as a finding: our oracle passed the causality guard, which is how we learned the competitor family was mis-specified (Section V).

We record two calibration facts. Divergent runs are excluded pairwise and reported as counts, because a run that dies early contributes a truncated sum and "wins" computation by dying. And a contrast that moves from $t = +17.9$ to $t = +5.4$ when the rival's tuning grid is densified, or from significant to nothing when a guard is enforced, is a statement about the protocol, not the mechanism; we report both values wherever that happened.

IX. Conclusions

We set out to show that a homeostatic wave field is a good distributed governor for the meta-parameters of MPC, and it is not: in every physically grounded arena it is beaten or matched by a minimal mechanism — and the margin by which we can bound the ties is stated with every claim.

What replaces it is a design lesson with two instances. Where a disturbance travels, deliver its warning with *constant* anticipation: an affine delay line, four parameters, statistically

TABLE V
EXECUTABLE AUDIT SUITE: 21 CHECKS OVER 18 ASSERTIONS (A9 RUNS AT THREE DAMPING RATIOS). EACH ENCODES A DEFECT FOUND AT LEAST ONCE.

id	assertion
A1–A3	terminal inequality, invariance of Ω, constraint satisfaction
A4–A5	$V^\star(x,\cdot)$ concave in α; admissible set can be non-convex
A6	the certified radius is valid
A7	first/second-order matching is exact *in discrete time*
A8	anti-windup does not fire when nothing is blocked
A9	executed overshoot equals that of the *discrete* system
A10–A11	gyroscopic collocation stable; Merkin threshold
A12	the inherited terminal ingredient fails (documented regression)
A13–A14	monotonicity duality of V_N in N
A15	scale invariance: normalized cost *curves* coincide; fails on boundary optima (Prop. 1)
A16	anticipation unusable without storage (viability filter)
A17	the field with $\beta = 0$ is exactly LTI (Prop. 2)
A18	DC cross-gain of the two forcings (Prop. 3)

at the clairvoyant ceiling, worth 12.4% against the best local detection, with the proportional family that per-hop designs default to significantly and explicably worse. Where a horizon must be rationed, react to the one event in the suboptimality signal that theory distinguishes — the certificate breaking — through a slow leak: two effective parameters, below the fixed-horizon frontier in a single loop, 5–6% ahead of the best fixed horizon in a fleet at equal spent computation, robustly to disturbance correlation, with the multiplexing gain correctly attributed to the allocation component alone.

The structural limits say these are not accidents of tuning: nothing built on absolute amplitude can carry information to a meta-parameter, an LTI field offers reshaping rather than information, its natural forcing carries nothing sustained at all, and quantization caps its reach logarithmically. And the five leaks say something about practice that we did not expect to be the paper's most durable contribution when we began: every one of our arenas was first decided by a defect, the defects were ordinary, the guards are cheap, and two of the five pushed the result *against* the mechanism we were defending. Honest comparisons are not a matter of intention; they are a matter of instrumentation.

REPRODUCIBILITY

All experiments, figures and the audit suite are generated by archived scripts; a build-time check verifies every number in this manuscript against the corresponding result file. Tuning and evaluation seeds are disjoint in every experiment reported.

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